

5.1. Tensor product $v \otimes w = ?$

a) $v(1, -1), w(2, 0)$

$$v \otimes w = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ -2 & 0 \end{bmatrix}$$

b) $v(1, 0, -1), w(0, 1, 1)$

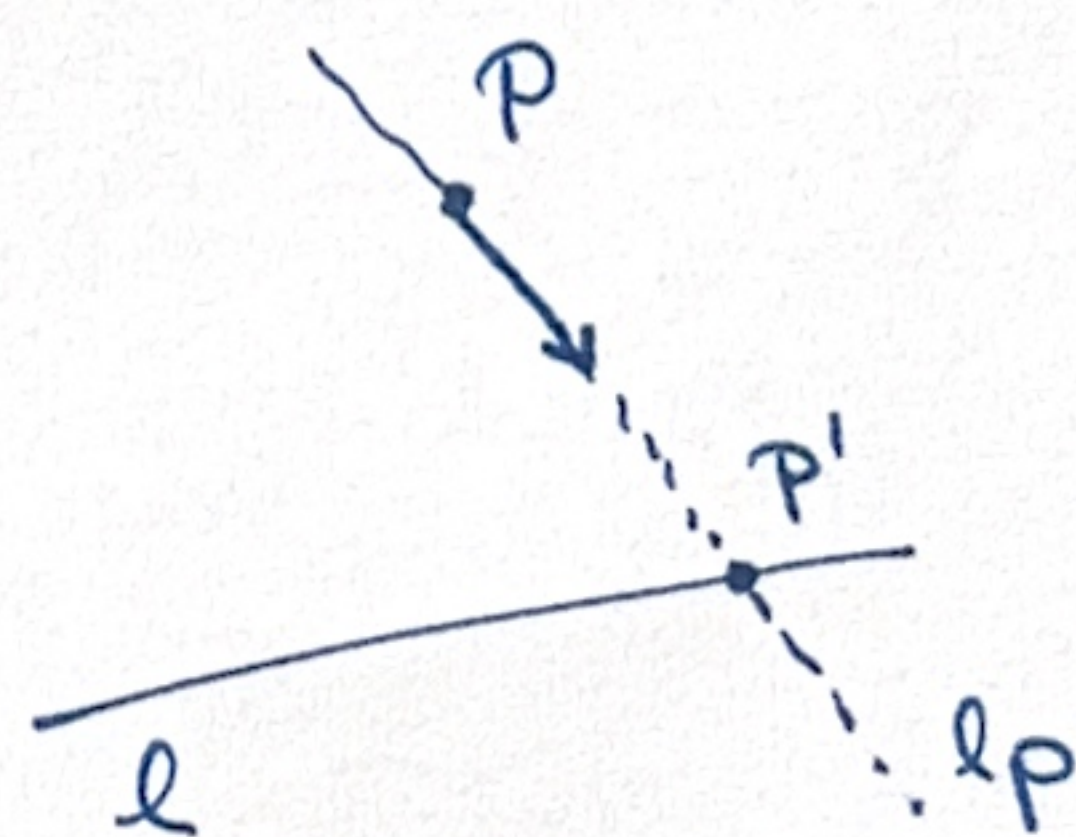
$$v \otimes w = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & -1 & -1 \end{bmatrix}$$

5.2. line $l: x + y - 1 = 0$

matrix forms of the -parallel projection in the line $l \parallel v(2, 1)$.
-parallel reflection

- (M1):
- take $P(x_P, y_P)$ arbitrary point.
 - construct line through P in direction of v .

$$l_P: \begin{cases} x = x_P + 2t \\ y = y_P + t \end{cases}$$



- intersect l with l_P to obtain the projection P' of P .

$$l \cap l_P: \begin{cases} x = x_P + 2t \\ y = y_P + t \\ x + y - 1 = 0 \end{cases}$$

$$\frac{x_P - 2y_P + 2}{3}$$

$$\frac{2y_P - x_P + 1}{3}$$

$$(x_P + 2t) + (y_P + t) - 1 = 0$$

$$\Rightarrow t = \frac{1 - x_P - y_P}{3}$$

$$\Rightarrow P' = l \cap l_P \text{ has coordinates } \left(x_P + 2 \cdot \frac{1 - x_P - y_P}{3}, y_P + \frac{1 - x_P - y_P}{3} \right)$$

- separate the coordinates of the generic point to obtain

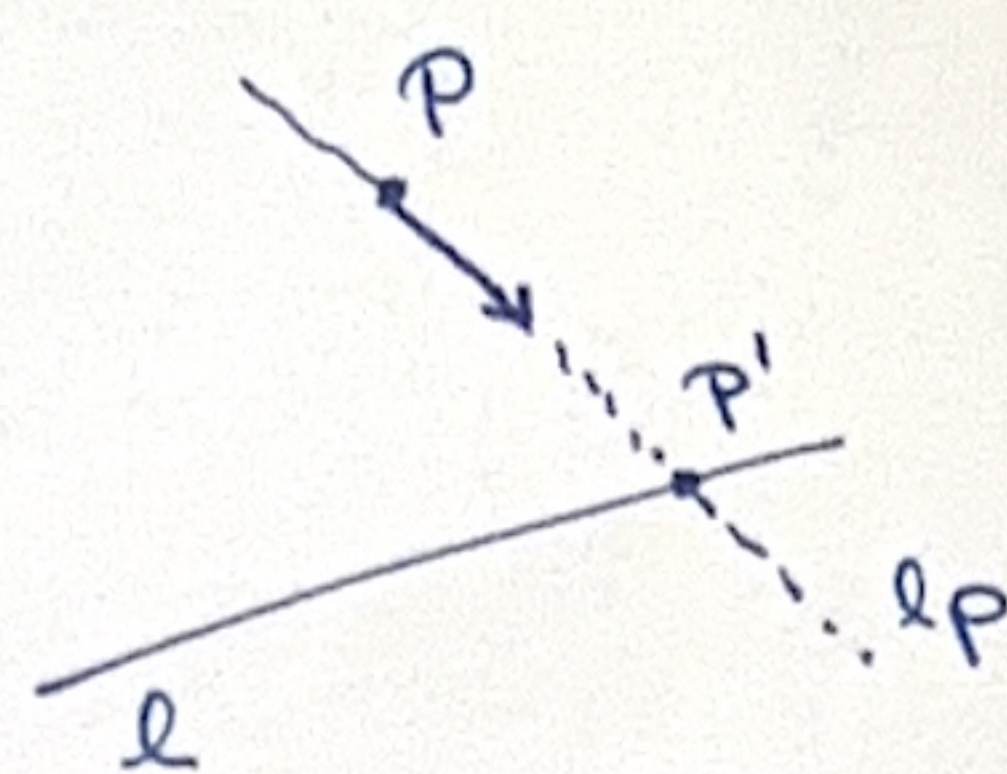
$$\begin{bmatrix} x_P \\ y_P \end{bmatrix} \mapsto \frac{1}{3} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_P \\ y_P \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

(H2). Apply formula for the projection.

$$Pr_{l,v}(P) = \left(I_2 - \frac{v \cdot a^T}{v^T \cdot a} \right) P - \frac{a_3}{v^T \cdot a} v$$

b) $v(1,0,-1), w(0,1,1)$
 $v \otimes w = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & -1 & -1 \end{bmatrix}$

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$$l \cap l_p: \begin{cases} x = x_p + 2t \\ y = y_p + t \\ x + y - 1 = 0 \end{cases}$$

$$\frac{x_p - 2y_p + 2}{3}$$

$$\frac{2y_p - x_p + 1}{3}$$

$$(x_p + 2t) + (y_p + t) - 1 = 0$$

$$\Rightarrow t = \frac{1 - x_p - y_p}{3}$$

$\Rightarrow P' = l \cap l_p$ has coordinates $(x_p + 2 \cdot \frac{1 - x_p - y_p}{3}, y_p + \frac{1 - x_p - y_p}{3})$

Seminar 5 warm-up

5.3. orthogonal proj. of $A(2, 11, -5)$ on $\pi: x + 4y - 3z + 7 = 0 \rightarrow$ matrix form.
 $m = (1, 4, -3), a_{m+1} = 7, a = m$

$$P_{\pi}^{\perp}(A) = A - \frac{a^T A + a_{m+1}}{|a|^2} \cdot a$$

$$a^T A + 7 = (1 \ 4 \ -3) \begin{pmatrix} 2 \\ 11 \\ -5 \end{pmatrix} + 7 = 2 + 44 + 15 + 7 = 24 + 44 = 68$$

$$|a|^2 = |m|^2 = (\sqrt{1^2 + 4^2 + 3^2})^2 = \sqrt{26} = 26$$

$$P_{\pi} = \begin{pmatrix} 2 \\ 11 \\ -5 \end{pmatrix} - \frac{68}{26} \cdot \begin{pmatrix} 1 \\ 4 \\ -3 \end{pmatrix} = \begin{pmatrix} 2 - 34 \\ 11 - 4 \cdot 34 \\ -5 - 34 \cdot (-3) \end{pmatrix} = \begin{pmatrix} -32 \\ -125 \\ 97 \end{pmatrix}$$

$$P_{\pi} = \begin{pmatrix} 2 - \frac{34}{13} \\ 11 - \frac{4 \cdot 34}{13} \\ -5 + \frac{34}{13} \end{pmatrix} = \begin{pmatrix} -\frac{8}{13} \\ \frac{7}{13} \\ \frac{37}{13} \end{pmatrix}$$

$$x \mapsto \left(I - \frac{a \otimes a}{|a|^2} \right) x - \frac{a_{m+1}}{|a|^2} \cdot a$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \left(I - \frac{1}{26} \begin{pmatrix} 1 & 4 & -3 \\ 4 & 16 & -12 \\ -3 & -12 & 9 \end{pmatrix} \right) \begin{pmatrix} x \\ y \\ z \end{pmatrix} - \frac{7}{26} \begin{pmatrix} 1 \\ 4 \\ -3 \end{pmatrix}$$

5.4. orthogonal reflection of $P(6, -5, 5)$ in $\pi: 2x - 3y + z - 4 = 0 \rightarrow$ matrix.

$$Ref_{\pi}^{\perp}(P) = A - 2 \cdot \frac{a^T A + a_{m+1}}{|a|^2} \cdot a$$

$$a = (2, -3, 1), a_{m+1} = -4$$

$$a^T A = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \begin{pmatrix} 6 & -5 & 5 \end{pmatrix} - 4 = (2 \ -3 \ 1) \begin{pmatrix} 6 \\ -5 \\ 5 \end{pmatrix} - 4 = 12 + 15 + 5 - 4 = 27 + 1 = 28$$

$$|a|^2 = (\sqrt{2^2 + 3^2 + 1^2})^2 = (\sqrt{14})^2 = 14$$

$$Ref = \begin{pmatrix} 6 \\ -5 \\ 5 \end{pmatrix} - 2 \cdot \frac{28}{14} \cdot \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -5 \\ 5 \end{pmatrix} - \begin{pmatrix} 8 \\ -12 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ 7 \\ 1 \end{pmatrix}$$

$$x \mapsto \left(I - 2 \cdot \frac{a \otimes a}{|a|^2} \right) x - 2 \cdot \frac{a_{m+1}}{|a|^2} \cdot a$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto I - \frac{2}{14} \cdot \dots$$

5.6. $H: a_1x + a_2y + a_3z + a_4 = 0.$

$\mathbb{R}^3$

$$a = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$v \perp H \Rightarrow v^T a = a_1 v_1 + a_2 v_2 + a_3 v_3 \neq 0.$$

Parallel proj.

$$Pr_{H,v}(P) = \left(I_3 - \frac{v \cdot a^T}{v^T a} \right) P - \frac{a_4}{v^T a} \cdot v$$

Explicit matrix: $v^T a = (v_1 \ v_2 \ v_3) \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = v_1 a_1 + v_2 a_2 + v_3 a_3$

$$v \cdot a^T = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} (a_1 \ a_2 \ a_3) = \begin{pmatrix} v_1 a_1 & v_1 a_2 & v_1 a_3 \\ v_2 a_1 & v_2 a_2 & v_2 a_3 \\ v_3 a_1 & v_3 a_2 & v_3 a_3 \end{pmatrix}$$

$$Pr_{H,v}(P) = \left[I_3 - \frac{1}{v_1 a_1 + v_2 a_2 + v_3 a_3} \cdot \begin{pmatrix} v_1 a_1 & v_1 a_2 & v_1 a_3 \\ v_2 a_1 & v_2 a_2 & v_2 a_3 \\ v_3 a_1 & v_3 a_2 & v_3 a_3 \end{pmatrix} \right] P - \frac{a_4}{v_1 a_1 + v_2 a_2 + v_3 a_3} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

Parallel refl.

$$Ref_{H,v}(P) = \left(I_3 - 2 \cdot \frac{v \cdot a^T}{v^T a} \right) P - 2 \cdot \frac{a_4}{v^T a} \cdot v.$$

If plane passes through origin $\Rightarrow a_4 = 0$, $\Rightarrow$ no translation part.

5.7. Matrix form $\rightarrow$ orthog. refl. in $\Pi_1: 3x - 4z = -1$, $\Pi_2: 10x - 2y + 3z = 4$

$$Ref_{\Pi}^{\perp}(x) = \left(I - 2 \cdot \frac{m \otimes m}{|m|^2} \right) x - 2 \cdot \frac{a_{n+1}}{|m|^2} \cdot m$$

$$\Pi_1: 3x - 4z + 1 = 0$$

$$\Rightarrow m_1 = (3, 0, -4), c = 1.$$

$$m \otimes m = m \cdot m^T = \begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix} (3 \ 0 \ -4) = \begin{pmatrix} 9 & 0 & -12 \\ 0 & 0 & 0 \\ -12 & 0 & 16 \end{pmatrix}$$

$$|m_1|^2 = \sqrt{25}^2 = 25.$$

$$Ref = \left(I - 2 \cdot \frac{1}{25} \begin{pmatrix} 9 & 0 & -12 \\ 0 & 0 & 0 \\ -12 & 0 & 16 \end{pmatrix} \right) x - 2 \cdot \frac{1}{25} \cdot \begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$$

$$= \left[\frac{1}{25} \begin{pmatrix} 25 & 0 & 0 \\ 0 & 25 & 0 \\ 0 & 0 & 25 \end{pmatrix} - \frac{1}{25} \begin{pmatrix} 18 & 0 & -24 \\ 0 & 0 & 0 \\ -24 & 0 & 36 \end{pmatrix} \right] \cdot x - \frac{1}{25} \begin{pmatrix} 6 \\ 0 \\ -8 \end{pmatrix}$$

= ...

$$\Pi_2: 10x - 2y + 3z - 4 = 0$$

$$\Rightarrow m_2 = (10, -2, 3), c = -4$$

$$m \otimes m = m \cdot m^T = \dots$$

$$|m_2|^2 = 113$$

$$Ref = \left(I - 2 \cdot \frac{1}{113} \begin{pmatrix} 100 & -20 & 30 \\ -20 & 4 & -6 \\ 30 & -6 & 9 \end{pmatrix} \right) x - 2 \cdot \frac{-4}{113} \begin{pmatrix} 10 \\ -2 \\ 3 \end{pmatrix} \sim \dots$$

Fixed points of the composition of the two refl. $\Rightarrow \pi_1 \cap \pi_2$.

$$\begin{cases} 3x - 4z = -1 & \Rightarrow \cancel{3x + 1 = 4z} \\ 10x - 2y + 3z = 4 & \underline{z = t} \end{cases}$$

$$3x - 4t = -1 \Rightarrow 3x = 4t - 1 \Rightarrow x = \frac{4t - 1}{3}$$

$$10 \cdot \frac{4t - 1}{3} - 2y + 3t = 4 \quad | \cdot 3$$

$$10(4t - 1) - 6y + 9t - 12 = 0$$

$$40t - 10 - 6y + 9t - 12 = 0$$

$$49t - 6y = 22$$

$$6y = 49t - 22$$

$$y = \frac{49t - 22}{6}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{4t - 1}{3} \\ \frac{49t - 22}{6} \\ t \end{bmatrix} \quad \text{§}$$

